WaterTAP Costing Reference Guide

Detailed Unit Model Costing Parameters

These models compute costs from physics-derived sizing variables. Capital cost equations vary by unit type. All parameters are tunable.

Unit Model	Capital Cost Eq.	Key Default Parameters	Reference
Anaerobic Digester	$A \times (Q/Q_{ref})^B$	A = \$19.36M B = 0.6 Q_{ref} = 911 m^3/day	NREL Waste-to-Energy Model
CSTR	$A \times V^B$	A = \$1,246.1/m^3 B = 0.71	C.C. Tang, 1984
Chiller	$C_{chill} \times (W_{duty}/COP)$	C_{chill} = \$200/kW COP = 7	
Clarifier (Circular)	$A \times A_{surf}^2 + B \times A_{surf} + C$	A = -6×10^{-4} B = 98.952 C = 191,806	Sharma et al. 2013
Clarifier (Primary)	$A \times Q_{MGD}^B$	A = \$538,746 B = 0.7	Byun et al. 2022
Clarifier (Rectangular)	$A \times A_{surf}^2 + B \times A_{surf} + C$	A = -2.9×10^{-3} B = 169.19 C = \$94,365	Sharma et al. 2013
Compressor	$C_{comp} \times W_{mech}^B$	C_{comp} = \$7364 B = 0.7	El-Sayed et al., 2001
Crystallizer	$A \times (Q/Q_{ref})^B \times IEC$	A = \$675,000 Q_{ref} = 1 m^3/hr B = 0.53 IEC = 1.43 steam = \$0.004/kg	Woods, 2007; Diab and Gerogiorgis, 2017; Yusuf et al., 2019; Panagopoulos (2019)
Dewatering (Centrifuge)	$A \times Q + B$	A = \$328.03/(gal/hr) B = \$751,295	McGivney & Kawamura, 2008
Dewatering (Filter Belt Press)	$A \times Q + B$	A = \$146.29/(gal/hr) B = \$433,972	McGivney & Kawamura, 2008
Dewatering (Filter Plate Press)	$A \times Q^B$	A = \$102,784/(gal/hr) B = 0.4216	McGivney & Kawamura, 2008
Electrodialysis	$C_{mem} \times A_{mem} + C_{electrode}$	C_{mem} = \$160/m^2 $C_{electrode}$ = \$2100/m^2 replacement = 20 %/yr (mem + electrode)	
Electrolyzer	$C_{mem} + C_{anode} + C_{cathode}$	C_{mem} = \$25/m^2 C_{anode} = \$300/m^2 $C_{cathode}$ = \$600/m^2 mat fraction = 65 %	Desalination 452 (2019) 265–278
Energy Recovery Device	$C_{ERD} \times Q$	C_{ERD} = \$535/(m^3/s)	
Evaporator	$C_{evap} \times A_{evap}$	C_{evap} = \$1000/m^2 material factor = 1.0	
GAC	$C_{contactor} + C_{gac} + C_{other}$	$C_{contactor}$ = polynomial f(vol) C_{gac} = exponential f(mass) C_{other} = power f(vol) C_{regen} = \$4.28/kg C_{makeup} = \$4.58/kg	EPA-WBS 2021
Heat Exchanger	$C_{HX} \times A_{hx}$	C_{HX} = \$300/m^2 steam = \$0.008/kg	

Heater (Electric)	$C_{heat} \times W_{duty}/eff$	C_{heat} = \$66/kW eff = 0.99	
Ion Exchange (Cation / Anion)	$C_{resin} + C_{vessels} + C_{tanks}$	$C_{res,AX}$ = \$205/ft^3 $C_{res,CX}$ = \$153/ft^3 vessel A=1596.5, b=0.46 resin replacement = 5 %/yr	EPA-WBS 2021
Membrane Distillation	$C_{mem} \times A_{mem}$	C_{mem} = \$56/m^2 replacement = 20 %/yr	
Mixer	$C_{mix} \times Q$	Generic = \$361/(L/s) NaOCl mixer = \$5.08/(m^3/day) $CaOH_2$ = 873.9/(kg/day)	
Nanofiltration	$C_{mem} \times A_{mem}$	C_{mem} = \$15/m^2 replacement = 20 %/yr	
OARO	$C_{mem} \times A_{mem}$	C_{mem} = \$30/m^2 $C_{mem,HP}$ = \$50/m^2 replacement = 15 %/yr	Bartholomew et al. 2017
Pressure Exchanger	$C_{PX} \times Q$	C_{PX} = \$535/(m^3/s)	
Pump (high pressure)	$C_{pump} \times W_{mech}$	C_{pump} = \$53/W	Malek et al. 1996
Pump (low pressure)	$C_{pump} \times Q$	C_{pump} = \$889/(m^3/s)	
Reverse Osmosis	$C_{mem} \times A_{mem}$	C_{mem} = \$30/m^2 $C_{mem,HP}$ = \$75/m^2 replacement = 20 %/yr	Bartholomew et al. 2018
Steam Ejector	$A \times (M)^B$	M = motive steam + entrained vapor; kg/hr A = \$1949 B = 0.3 steam = \$0.008/kg	Gabriel 2015, Desalination
Thickener	$A \times A_{surf} + B$	A = 4729.8/ft^2 B = \$37,068	McGivney & Kawamura, 2008
UV+AOP	$C_{reactor} + C_{lamp}$	$C_{reactor}$ = \$202.35/kW C_{lamp} = \$235.50/kW lamp replacement = 33.3 %/yr	

WaterTAP Costing Reference Guide (continued)

Zero-Order Unit Model Costing — Default Parameters

Zero-order models use: $C_{cap} = A \times (Q_{in}/Q_{basis})^B$. A is in USD for the reference year shown. Costs are CPI-adjusted to the study year.

Unit Model	A	B	Q_{basis}	Cost Year	Energy (kWh/m³)	Recovery	Reference
Aeration Basin	725,570	0.586	1 MGD	2014	0.412	N/A	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Air Flotation	24,348	0.867	1 MGD	2007	0.3	0.9999	McGivney & Kawamura Figure 5.5.36b
Anaerobic Digestion Oxidation	19,355,231	0.600	911 m³/hr	2012	0.15	0.9999	NREL Waste-to-Energy Model (WESyS)
Anaerobic Digestion Reactive	19,355,231	0.600	911 m³/hr	2012	0.2	N/A	Unknown
Backwash Solids Handling	2,870,106	0.918	100 MGD	2007	f(Q)	0.95	Based on costs for subprocesses in Table 5.7.1 (Equations 29, 30, 31, 32, 33, 37, 38, 39) in Cost Estimating Manual for Water Treatment Facilities (McGivney/Kawamura)
Bio-Active Filtration	725,570	0.586	1 MGD	2014	None	0.99	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Bioreactor	1,252,160	1	45 m³/hr	2018	None	0.97	Developed with Aspen Cost Estimation for San Luis Case Study in WT3.
Blending Reservoir	2,000,000	0.700	50,000 m³/hr	2020	None	N/A	Unknown
Buffer Tank	1,000,000	0.700	29,526 m³/hr	2019	None	N/A	Unknown
CO_2 Addition	464,000	0.700	1 MGD	2019	None	N/A	Unknown
Cartridge Filtration	725,570	0.586	1 MGD	2014	0.0002	0.9999	Texas Water Development Board IT3P Figure 3.3
Clarifier	177,516	0.318	5,450 m³/hr	2007	None	0.999	McGivney & Kawamura, Figure 5.5.24c, Cost Estimating Manual for Water Treatment Facilities
Conventional Activated Sludge	725,570	0.586	1 MGD	2014	0.14	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Cooling Supply	40,299	0.866	1 MGD	2007	None	N/A	McGivney & Kawamura Figure 5.5.35b
Cooling Tower	0	1	1 m³/hr	2020	None	0.25	Unknown
Decarbonator	150,000	1	1,577 m³/hr	2007	None	N/A	Survey of High-Recovery and Zero Liquid Discharge Technologies for Water Utilities (2008). Table A2.2.
Dissolved Air Flotation	336,873	0.318	911 m³/hr	2007	0.05	0.9999	Unknown
Dual Media Filtration	3,481,852	0.586	30 MGD	2007	0.0507	0.99	Unknown
Electrodialysis Reversal	31,000,000	1	946 m³/hr	2016	f(x)	0.9214	Unknown
Energy Recovery	100,000	0.700	4 m³/hr	2020	-3.17	N/A	Unknown
Feed Water Tank	2,000,000	0.700	1,736 m³/hr	2018	None	N/A	Unknown
Fixed Bed Bioreactor	1,542,008	0.734	1 MGD	2017	0.0693	N/A	Regressed from standard designs for EPA-WBS Biological Treatment with Fixed Bed using pressure vessels from 2017
Injection Well Disposal	22,500,000	0.700	473 m³/hr	2011	0.41	N/A	Unknown
Intrusion Mitigation	44,600,000	1	4,750 m³/hr	1998	0.0512	N/A	Monterey One Case Study Data
Landfill	1,000,000	0.700	1 m³/hr	2020	None	N/A	Unknown
Media Filtration	725,570	0.586	1 MGD	2014	0.00015	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Membrane Bioreactor	725,570	0.586	1 MGD	2014	f(Q)	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Microfiltration	500,000	1	1 MGD	2014	0.18	0.95	Texas Water Development Board IT3PR Table 3.21
Municipal Drinking	40,299	0.866	1 MGD	2007	f(Q)	N/A	McGivney & Kawamura Figure 5.5.35b
Municipal Wwtp	1,000,000	0.700	100,000 kg/hr	2020	None	N/A	Unknown
Nanofiltration	750,000	1	1 MGD	2014	0.231	0.85	Texas Water Development Board IT3P Table 3.21

Primary Separator	603,553	0.318	5,450 m³/hr	2007	None	0.999	McGivney & Kawamura, Figure 5.5.24c, Cost Estimating Manual for Water Treatment Facilities
Pump	40,299	0.866	1 MGD	2007	0.051	N/A	McGivney & Kawamura Figure 5.5.35b
SMP	1,000,000	0.700	100,000,000 kg/hr	2020	None	N/A	Unknown
Screen	11	0.965	1 m³/d	2018	None	0.9999	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management. Figure 4.6
Seawater Intake	3,624	0.889	1 m³/hr	2018	f(Q)	N/A	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management, Figure 4.2 + Figure 4.4
Secondary Treatment WWTP	1,000,000	0.700	100,000 kg/hr	2020	0.14	N/A	Unknown
Settling Pond	177,516	0.318	5,450 m³/hr	2007	None	0.9999	McGivney & Kawamura, Figure 5.5.24c, Cost Estimating Manual for Water Treatment Facilities
Sludge Tank	1,000,000	0.700	100,000 kg/hr	2020	None	0.99999	Unknown
Static Mixer	1,698	0.325	1 L/s	2010	None	N/A	Chemical Engineering Design, 2nd Edition. Principles, Practice and Economics of Plant and Process Design. Table 7.2 eq fit to power eq.
Surface Discharge	35,000,000	0.873	10,417 m³/hr	2020	f(Q)	N/A	Unknown
Tramp Oil Tank	1,000,000	0.700	100,000 kg/hr	2020	None	N/A	Unknown
Tri Media Filtration	725,570	0.586	1 MGD	2014	0.00045	0.9	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Ultrafiltration	500,000	1	1 MGD	2014	0.231	0.95	Texas Water Development Board IT3P Table 3.21
WAIV	110,000	1	64 kg/hr	2020	0.65	0.0001	Unknown
Walnut Shell Filter	725,570	0.586	1 MGD	2014	None	0.99	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Water Pumping Station	24,348	0.867	1 MGD	2007	None	N/A	McGivney & Kawamura Figure 5.5.36b

All parameters are defaults and fully tunable by the user. Source: watertap-org/watertap GitHub repository.